EXPLORATORY TESTING CHARTER & SESSION REPORT

Project Name	MediPlus Pharmacy Mobile App	**Session Duration**	90 Minutes (Timeboxed)
Client / Organization	MediPlus Pharmacy	**Session Date**	November 10, 2026
QA Engineer	Muhammad Huzaifa Waheed	**Build Under Test**	mda-2.3.0-27.apk (Sauce Labs)
Primary Device	Google Pixel 4 (Android 13 / API 33)	**Charter Focus**	Unexpected User Actions & Boundary Stress

CHARTER OVERVIEW

Mission & Objective

Explore the MediPlus Android mobile application beyond scripted test scenarios by simulating unpredictable, aggressive, and non-standard user behaviors. The objective is to identify potential functional leaks, client-side input validation weaknesses, payment gateway bypasses, and state-persistence issues.

Target Exploration Areas

1. **Input Stress & Character Injection:** Special characters, multi-byte UTF-8 symbols, emojis, and whitespace injection into shipping forms.
2. **Payment Boundary Validation:** Non-standard card lengths, Luhn algorithm compliance, and past expiration dates.
3. **Session State & Data Privacy:** Cart state behavior across login, logout, application kill, and background lifecycle transitions.

90-MINUTE TIME BOX ALLOCATION

Phase	Time Box	Focus & Activity
Phase 1	00m - 15m	**Setup & Baseline:** Verification of Pixel 4 (API 33), clean cache, baseline checkout verification.
Phase 2	15m - 45m	**Session 1 (Input Stress):** Entering extreme strings, emojis, special characters into Address & City fields.
Phase 3	45m - 70m	**Session 2 (Payment Gateway):** Testing expired card dates, irregular digit lengths, and negative CVV values.
Phase 4	70m - 85m	**Session 3 (State Persistence):** Testing cart session persistence during account logout and backgrounding.

Phase 5	85m - 90m	**Triage & Logging:** Reviewing captured evidence, cataloging defects BUG_006 to BUG_008.

SESSION EXECUTION NOTES (OBSERVATIONS & HEURISTICS)

- **Heuristic Applied – "Goldilocks & Boundary Tampering":** Tested with too short, too long, and unexpected character types across all single-line input controls.
- **Heuristic Applied – "Follow the Data":** Tracked user address entries from shipping details directly through to payment checkout and order summary.
- **Key Observations:**
    - **Input Sanitation Failure:** The shipping address form accepts emojis and symbols without client-side regex enforcement. Data propagates directly to subsequent checkout views without sanitation or warning dialogs.
    - **Payment Expiration Bypass:** The payment gateway allows expired month/year inputs (e.g., 03/25 during a 2026 test session) and enables the submission button instead of blocking the transaction at the client boundary.
    - **Card Length Tolerance:** Non-Visa digit sequences allow arbitrary character input lengths, bypassing standard 16-digit card validation blocks.
    - **Session Isolation Leak:** User logout invalidates the user authentication token but leaves the local cart database intact. A newly arriving guest or unauthenticated user inherits the preceding user's active shopping cart.
    - **UI Stability:** The app did not crash or trigger native ANR (Application Not Responding) dialogs under heavy character loads, demonstrating good thread handling despite validation logic shortcomings.

DEFECT LOG EXTENSION (NEW FINDINGS)

Bug 1: BUG_006

- **Title:** Emoji and Special Character Injection Allowed in Shipping Address Fields
- **Severity:** Medium | **Priority:** Medium
- **Steps to Reproduce:**
    1. Add any product to the cart and navigate to the Checkout Address screen.

2. Enter special characters and emojis in the Full Name, Address, and City fields (e.g., "Huzaifa😀#", "Station 😀$", "Islamabad😀$").
3. Enter valid zip and country, then tap "To Payment".

- **Observed Result:** Input is accepted without error; unsanitized strings carry directly over to the Payment screen.
- **Expected Result:** Fields should enforce strict alphanumeric regex and reject emoji/script inputs with a clear inline validation message.

Bug 2: BUG_007

- **Title:** Payment Gateway Accepts Past Expiration Dates and Irregular Card Lengths
- **Severity:** High | **Priority:** High
- **Steps to Reproduce:**
  1. Proceed to the Checkout Payment screen.
  2. Enter an unformatted/non-standard card number (e.g., starting with arbitrary prefixes).
  3. Enter an expired expiration date (e.g., "03/25" in year 2026) and a 3-digit CVV.
  4. Tap "Review Order".
- **Observed Result:** The app accepts the expired date and irregular card length, navigating successfully to the Review Order screen.
- **Expected Result:** The app must validate expiration dates against the current system date and block submission with an "Expired card" warning.

Bug 3: BUG_008

- **Title:** Active Shopping Cart Session Persists After Explicit Account Logout
- **Severity:** High | **Priority:** High
- **Steps to Reproduce:**
  1. Log in as an authenticated user (bod@example.com).
  2. Add multiple items (e.g., 4 items) to the cart.
  3. Open the side menu, tap "Log Out", and confirm logout.

4. Inspect the cart badge and tap the Cart icon from the unauthenticated catalog screen.

- **Observed Result:** The cart maintains all 4 previously added items instead of clearing the session state.
- **Expected Result:** Explicit user logout must purge active cart caches or decouple them from unauthenticated guest sessions to protect user privacy.

SUMMARY RECOMMENDATIONS

1. **Implement Regex Validation:** Restrict shipping address input fields to alphanumeric characters and standard punctuation. Disallow emoji/symbol injection.
2. **Enforce Payment Card Rules:** Integrate standard client-side Luhn validation, strict card-type length bounds (15/16 digits), and date logic requiring Expiration Date > Current Month/Year.
3. **Session Cache Clearance:** Tie local cart state to user authentication tokens. Ensure that invoking the logout procedure triggers a complete cache wipe of cart contents and shipping forms.